Used Device Data Analysis

PGP DSBA - Guided Project Report

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9-8-2024

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1. Description:- Project Predictive Modeling: Used Device Data Analysis:-

Business Context

Buying and selling used phones and tablets used to be something that happened on a handful of online marketplace sites. But the used and refurbished device market has grown considerably over the past decade,and a new IDC (International Data Corporation) forecast predicts that the used phone market would be worth$52.7bn by 2023 with a compound annual growth rate (CAGR) of 13.6% from 2018 to 2023. This growth canbe attributed to an uptick in demand for used phones and tablets that offer considerable savings comparedwith new models.

Refurbished and used devices continue to provide cost-effective alternatives to both consumers and businesses that are looking to save money when purchasing one. There are plenty of other benefitsassociated with the used device market. Used and refurbished devices can be sold with warranties and canalso be insured with proof of purchase. Third-party vendors/platforms, such as Verizon, Amazon, etc., provideattractive offers to customers for refurbished devices. Maximizing the longevity of devices through second-hand trade also reduces their environmental impact and helps in recycling and reducing waste. The impact ofthe COVID-19 outbreak may further boost this segment as consumers cut back on discretionary spendingand buy phones and tablets only for immediate needs.

Objective

The rising potential of this comparatively under-the-radar market fuels the need for an ML-based solution to develop a dynamic pricing strategy for used and refurbished devices. ReCell, a startup aiming to tap the potential in this market, has hired you as a data scientist. They want you to analyze the data provided and build a linear regression model to predict the price of a used phone/tablet and identify factors that significantly influence it.

2. Data Description

The data contains the different attributes of used/refurbished phones and tablets. The data was collected in the year 2021. The detailed data dictionary is given below.

2.1 Data Dictionary

brand_name: Name of manufacturing brand

os: OS on which the device runs

screen_size: Size of the screen in cm

4g: Whether 4G is available or not

5g: Whether 5G is available or not

main_camera_mp: Resolution of the rear camera in megapixels

selfie_camera_mp: Resolution of the front camera in megapixels

int_memory: Amount of internal memory (ROM) in GB

ram: Amount of RAM in GB

battery: Energy capacity of the device battery in mAh

weight: Weight of the device in grams

release_year: Year when the device model was released

days_used: Number of days the used/refurbished device has been used

normalized_new_price: Normalized price of a new device of the same model in euros

normalized_used_price: Normalized price of the used/refurbished device in euros

Important Note

The following questions need to be answered as a part of the EDA section of the project:

1. What does the distribution of normalized used device prices look like?

2. What percentage of the used device market is dominated by Android devices?

3. The amount of RAM is important for the smooth functioning of a device. How does the amount of RAM vary with the brand?

4. A large battery often increases a device's weight, making it feel uncomfortable in the hands. How does the weight vary for phones and tablets offering large batteries (more than 4500 mAh)?

5. Bigger screens are desirable for entertainment purposes as they offer a better viewing experience. How many phones and tablets are available across different brands with a screen size larger than 6 inches?

6. A lot of devices nowadays offer great selfie cameras, allowing us to capture our favourite moments with loved ones. What is the distribution of devices offering greater than 8MP selfie cameras across brands?

7. Which attributes are highly correlated with the normalized price of a used device?

Kindly ensure to thoroughly examine all variables of the data in EDA, going

Explore the dataset and extract insights using Exploratory Data Analysis

3. Data Overview

3.1 Displaying the first five rows of the dataset:-

- The data frame has 15 columns as mentioned in the Data Dictionary.
- Data in each row corresponds to the Used device.

	brand_name	os	screen_size	4g	5g	main_camera_mp	selfie_camera_mp	int_memory	ram	battery	weight	release_year	days_used	normalized_used_price	normalized_new_price
0	Honor	Android	14.50	yes	no	13.0	5.0	64.0	3.0	3020.0	146.0	2020	127	4.307572	4.715100
1	Honor	Android	17.30	yes	yes	13.0	16.0	128.0	8.0	4300.0	213.0	2020	325	5.162097	5.519018
2	Honor	Android	16.69	yes	yes	13.0	8.0	128.0	8.0	4200.0	213.0	2020	162	5.111084	5.884631
3	Honor	Android	25.50	yes	yes	13.0	8.0	64.0	6.0	7250.0	480.0	2020	345	5.135387	5.630961
4	Honor	Android	15.32	yes	no	13.0	8.0	64.0	3.0	5000.0	185.0	2020	293	4.389995	4.947837

Table 1 Displaying the first five rows of the dataset

- The dataset contains information about different Used phones/tablets.
- Many devices seem to have a 4g with 13 main_camera_mp only, which are Android.

3.2 Checking the shape of the dataset:-

- The dataset contains information (15 attributes) about 3454 used Phones/tablets.

3.3 Checking the data types of the columns for the dataset:-

- The column range index has 3454 entries (i.e No.of rows = 3454) which starts the number series from 0 to 3454.

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3454 entries, 0 to 3453
Data columns (total 15 columns):
 #   Column                 Non-Null Count  Dtype
---  ------                 --------------  -----
 0   brand_name             3454 non-null   object
 1   os                     3454 non-null   object
 2   screen_size            3454 non-null   float64
 3   4g                     3454 non-null   object
 4   5g                     3454 non-null   object
 5   main_camera_mp         3275 non-null   float64
 6   selfie_camera_mp       3452 non-null   float64
 7   int_memory             3450 non-null   float64
 8   ram                    3450 non-null   float64
 9   battery                3448 non-null   float64
 10  weight                 3447 non-null   float64
 11  release_year           3454 non-null   int64
 12  days_used              3454 non-null   int64
 13  normalized_used_price  3454 non-null   float64
 14  normalized_new_price   3454 non-null   float64
dtypes: float64(9), int64(2), object(4)
memory usage: 404.9+ KB
```

Table 2 Checking the data types of the columns for the dataset

- Data types: - Three types of data types are as follows float64(9) (float = decimal number), int64(2) (i.e. int = integer), object (4) (i.e. Other than number & decimal number).
- The table data occupy the memory usage: 404.9+ KB

3.4 Statistical summary of the dataset:-

- This table reflects the statistical summary of the data contains - Count, Unique, top, freq, Mean, Std (Standard deviation), Min (Minimum), Max (Maximum), and Percentages (25%, 50%, 75%).

	count	unique	top	freq	mean	std	min	25%	50%	75%	max
brand_name	3454	34	Others	502	NaN	NaN	NaN	NaN	NaN	NaN	NaN
os	3454	4	Android	3214	NaN	NaN	NaN	NaN	NaN	NaN	NaN
screen_size	3454.0	NaN	NaN	NaN	13.713115	3.80528	5.08	12.7	12.83	15.34	30.71
4g	3454	2	yes	2335	NaN	NaN	NaN	NaN	NaN	NaN	NaN
5g	3454	2	no	3302	NaN	NaN	NaN	NaN	NaN	NaN	NaN
main_camera_mp	3275.0	NaN	NaN	NaN	9.460208	4.815461	0.08	5.0	8.0	13.0	48.0
selfie_camera_mp	3452.0	NaN	NaN	NaN	6.554229	6.970372	0.0	2.0	5.0	8.0	32.0
int_memory	3450.0	NaN	NaN	NaN	54.573099	84.972371	0.01	16.0	32.0	64.0	1024.0
ram	3450.0	NaN	NaN	NaN	4.036122	1.365105	0.02	4.0	4.0	4.0	12.0
battery	3448.0	NaN	NaN	NaN	3133.402697	1299.682844	500.0	2100.0	3000.0	4000.0	9720.0
weight	3447.0	NaN	NaN	NaN	182.751871	88.413228	69.0	142.0	160.0	185.0	855.0
release_year	3454.0	NaN	NaN	NaN	2015.965258	2.298455	2013.0	2014.0	2015.5	2018.0	2020.0
days_used	3454.0	NaN	NaN	NaN	674.869716	248.580166	91.0	533.5	690.5	868.75	1094.0
normalized_used_price	3454.0	NaN	NaN	NaN	4.364712	0.588914	1.536867	4.033931	4.405133	4.7557	6.619433
normalized_new_price	3454.0	NaN	NaN	NaN	5.233107	0.683637	2.901422	4.790342	5.245892	5.673718	7.847841

Table 3 Statistical summary of the dataset

There are various statistical measurements of used phones/tablets for each category (i.e each column) as follows,

- Count: - total number of non-missing values of the used phones/tablets
- unique: Number of unique values (for object/categorical columns).
- top: Most frequent value (for object/categorical columns).
- freq: Frequency of the most frequent value (for object/categorical columns).
- mean: Mean of the values in each column (i.e Integer).
- std: Standard deviation of the values in each column (i.e Integer).
- min: Minimum value in each column (including both Int & Float)
- 25%: 25th percentile (first quartile).
- 50%: 50th percentile (median).
- 75%: 75th percentile (third quartile).
- max: Maximum value in each column (including both Int & Float)

3.5 Checking for duplicate values:-

- There are 0 (i.e. zero) duplicate values.

3.6 Checking for missing values:-

- The six columns have missing values (i.e. blank cells).

```
brand_name                 0
os                         0
screen_size                0
4g                         0
5g                         0
main_camera_mp           179
selfie_camera_mp           2
int_memory                 4
ram                        4
battery                    6
weight                     7
release_year               0
days_used                  0
normalized_used_price      0
normalized_new_price       0
dtype: int64
```

Table 4 Checking for missing values

Exploratory Data Analysis

- EDA has a key role in data analysis for the data science. It includes different types of analysis like Univariate analysis, Bivariate & Multivariate analysis which are explained in the below points.

4. Univariate Analysis:-

4.1 Normalized_used_price:-

It is to create a histogram_boxplot for 'normalized_new_price'.

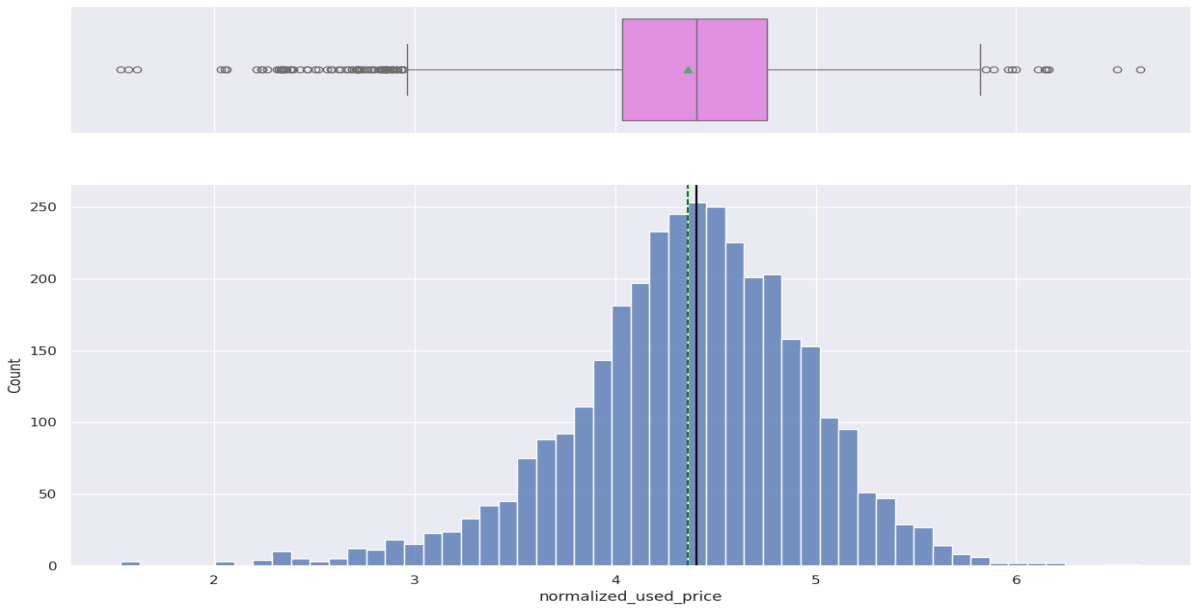

Figure 1 Normalized_used_price

- The normalized used prices are close to the normal distribution.
- The normalized used prices are distributed approximately 4.35 by used devices on average.
- It has 5 outliers of 2 on the right-most and 3 on the leftmost. It has occupied a small portion of the total price so it will give better results.
- Medium is 4.5 so it has normally distributed the used prices.

4.2 Normalized_new_price

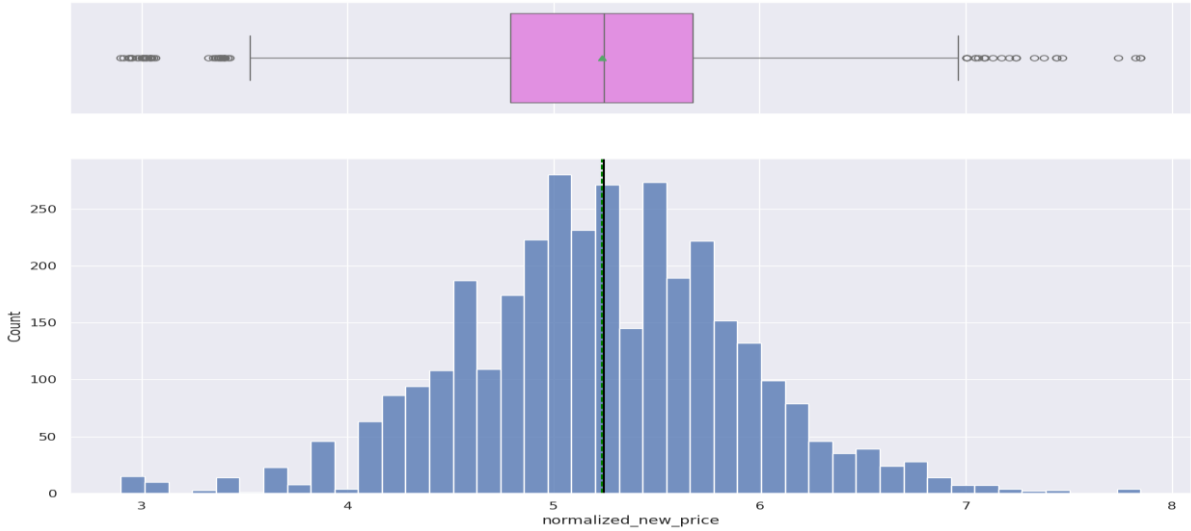

Figure 2 Normalized_new_price

- The normalized new prices are close to the normal distribution.
- The normalized used prices are distributed approximately 4.35 by used devices on average.
- It has many outliers of 3 on the right-most and many on the leftmost.
- Medium and mean both are similar approximately 5.20 so no difference in the normalized new prices of them.

4.3 screen_size:-

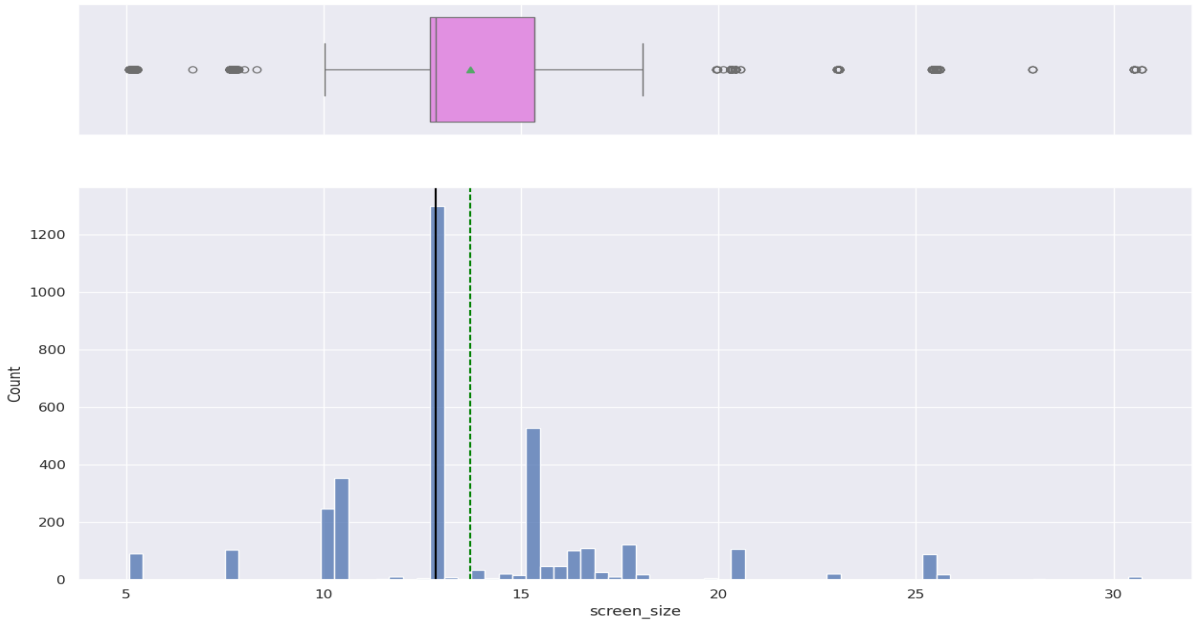

Figure 3 screen_size

- The screen size across all the brand devices is 13.70 (Approximately) inches on average.
- The median is 12.80 (approximately) which is less than the average.
- Boxplot has many outliers for screen size and it has right skewed.

4.4 main_camera_mp

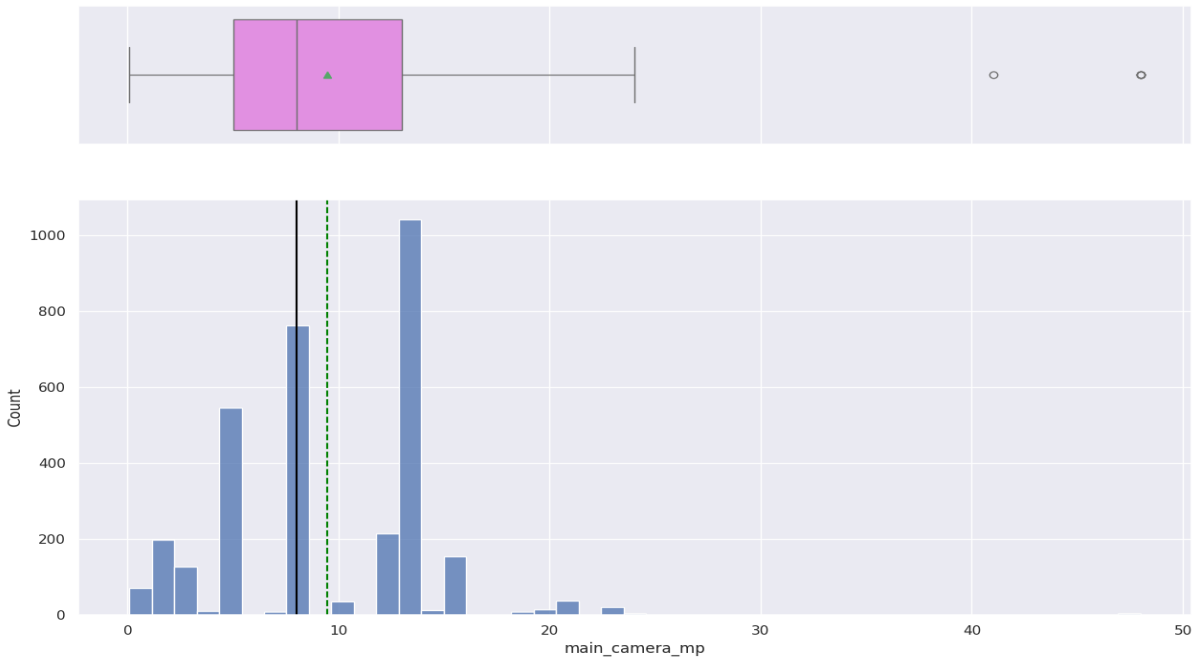

Figure 4 main_camera_mp

- The main_camera_mp across all the brand devices is 9.45 (Approximately) inches on average.
- The median is 9.0 (approximately) which is less than the average.
- Boxplot has 2 outliers on the right side so it is right skewed.

4.5 selfie_camera_mp:-

The below Histogram and boxplot have been prepared for the selfie camera mp details for each device.

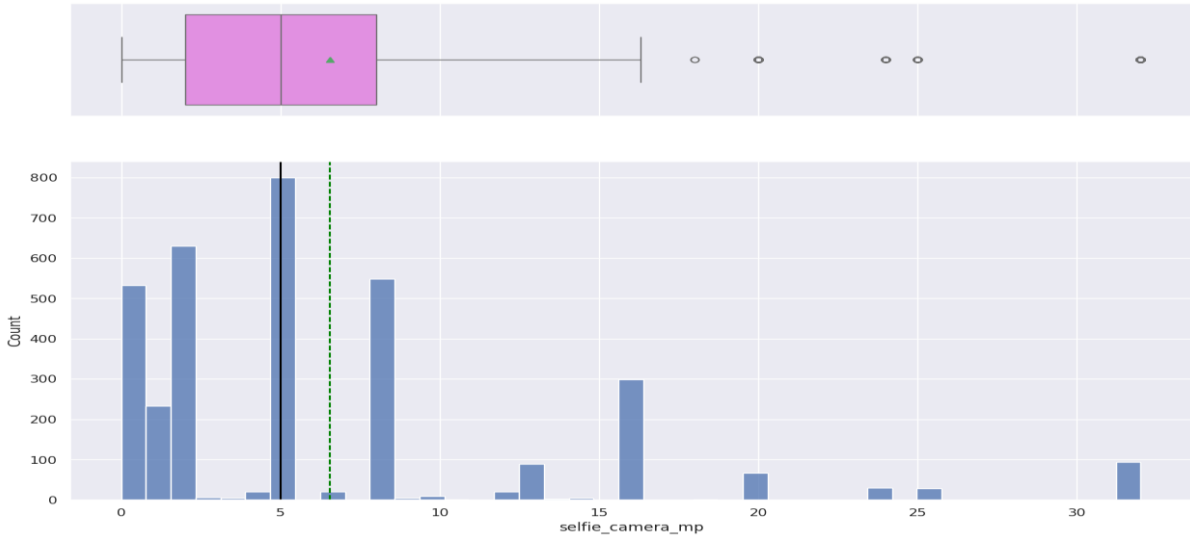

Figure 5 selfie_camera_mp

- It is right skewed.
- The 3rd quartile and maximum have more distance because of the 5 outliers on the right-hand side.
- Medium is 5 mp. Many devices contained 6.5 mp (approximately) on average.

4.6 int_memory:-

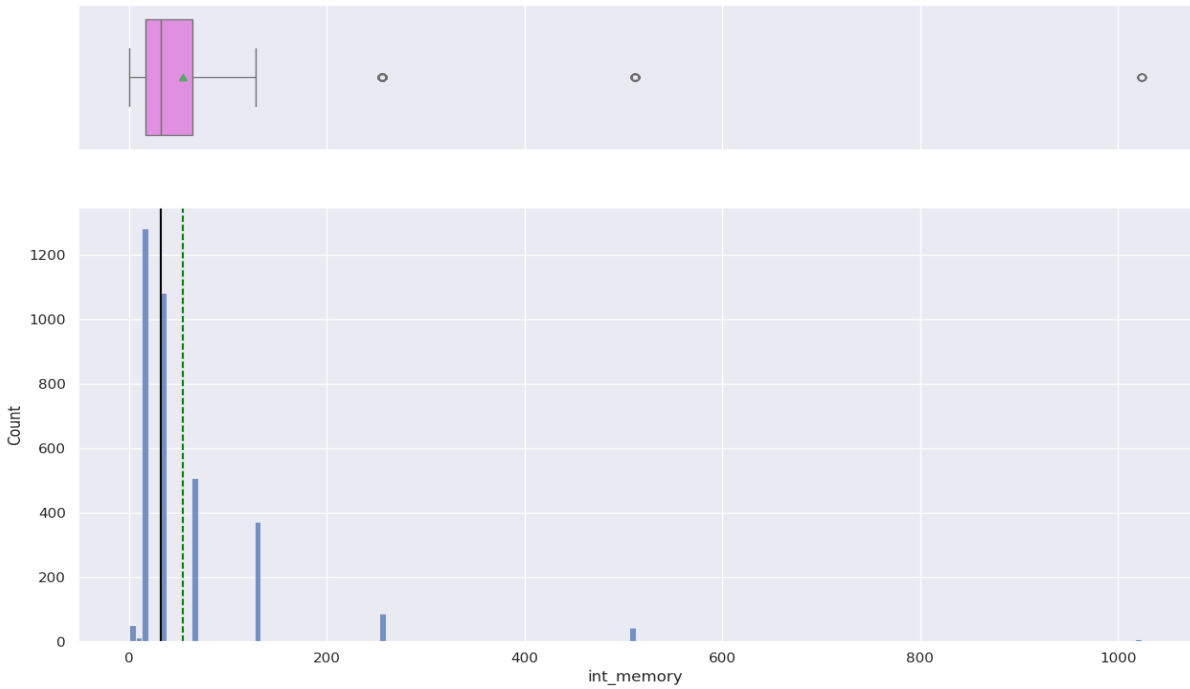

Figure 6 int_memory

- It is right skewed.

- The 1st quartile and minimum have low distance because of no outliers on the right-hand side.
- Medium is 32 GP which is lower than Average memory. Many devices use 54 GB (approximately) memory on average.

4.7 Ram:- RAM means Random Access Memory

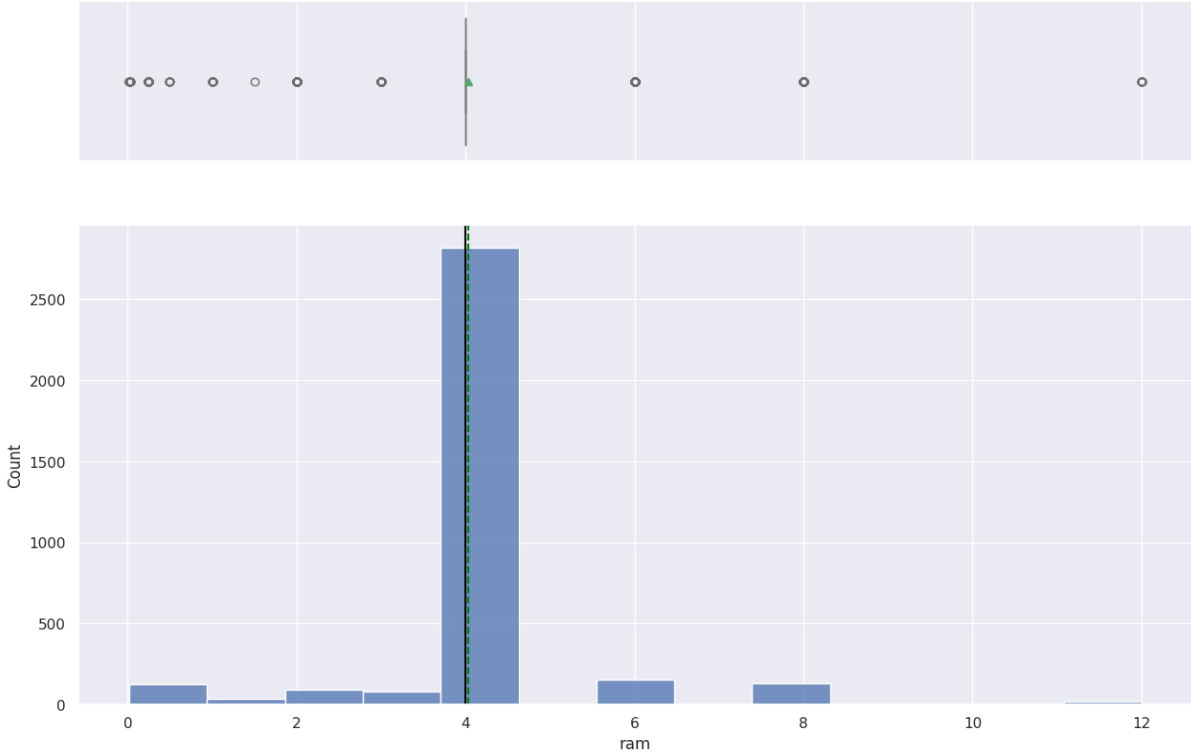

Figure 7 Ram

- It is right skewed.
- It has fewer outliers on the right side but it occupies more memory so it has more distance on the right-hand side.
- The Average use of RAM is 4 GB which is equal to Median.

4.8 Weight:-

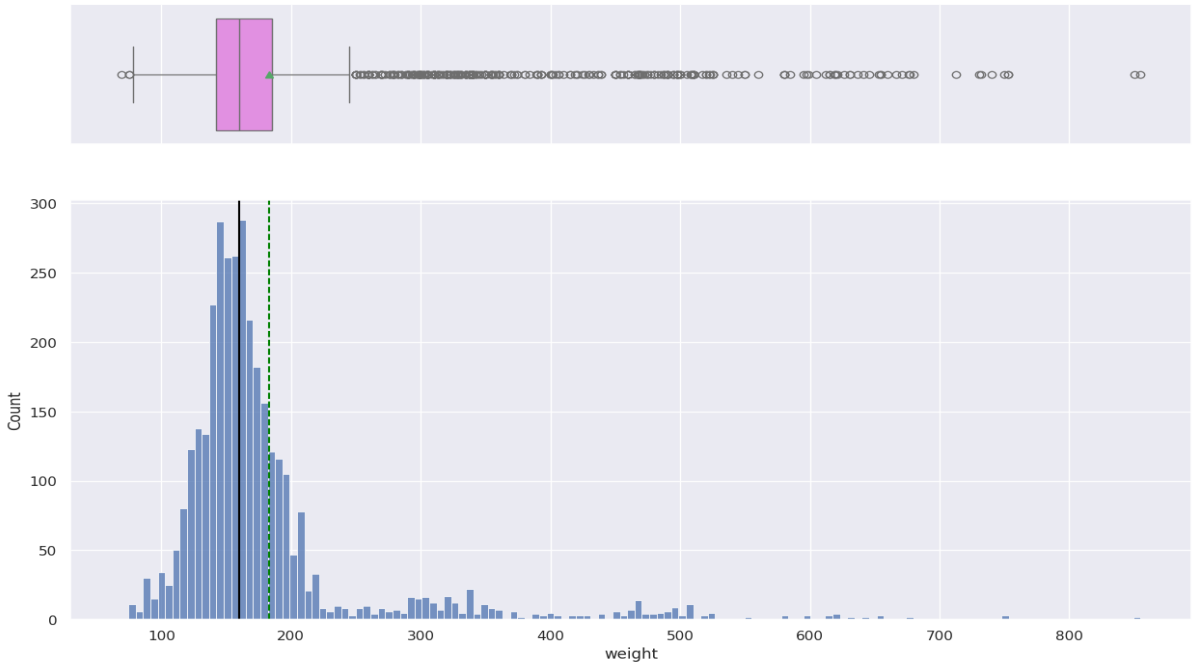

Figure 8 Weight

- It is right skewed.
- It has two outliers on the left side and many outliers on right-hand side.
- The Average weight of a device is 182 grams which is more than the Median of 160 grams.

4.9 Battery:-

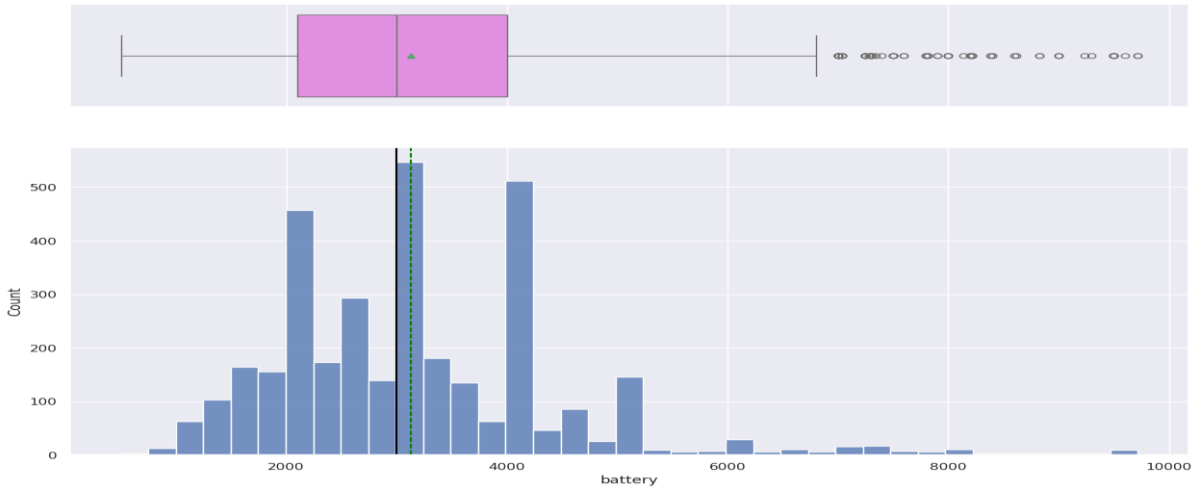

Figure 9 Battery

- It is right skewed.
- It has two outliers on the left side and many outliers on right-hand side.
- The Average weight of a device is 182 grams which is more than the Median of 160 grams.

4.10 days_used:- Number of days the used/refurbished device has been used

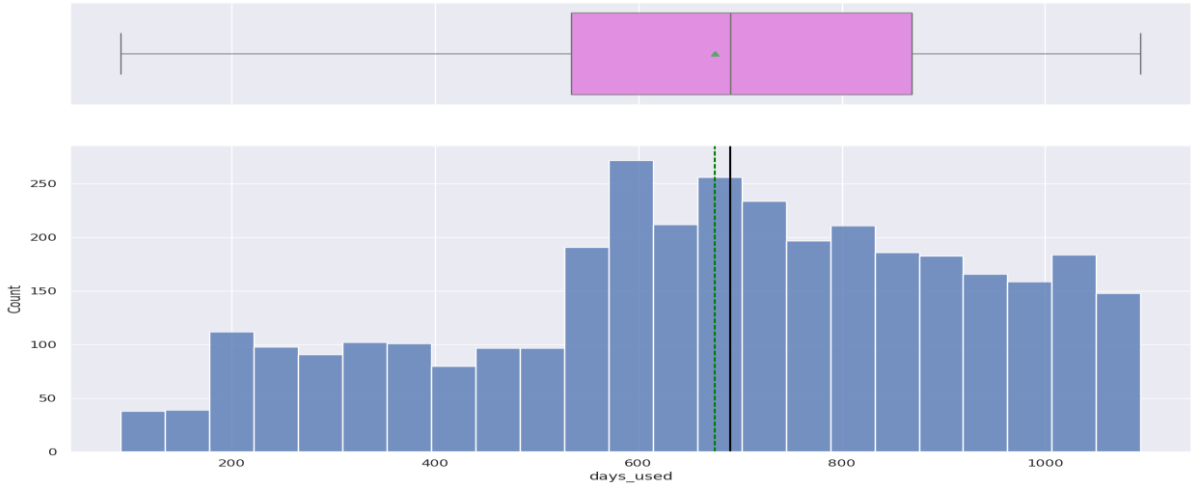

Figure 10 days_used

- It is left skewed.
- It has two outliers on the left side and many outliers on right-hand side.
- The average number of days used is 674 and the Median is 690.5 days.

4.11 brand_name:- The bar chart reflects 10 brands.

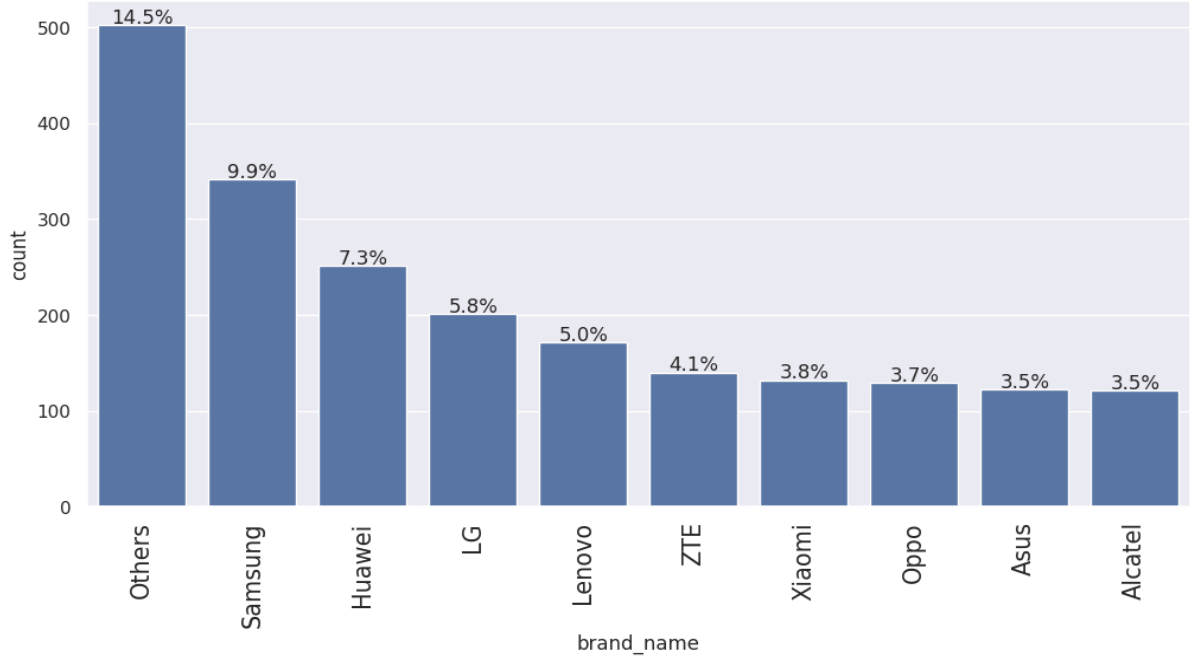

Figure 11 brand_name

- It gradually slopes to the right-hand side. Other brands occupy 14.5 percent of the market.
- Samsung is in second place with 9.9% and next is Huawei with 7.3%.
- Alcatel & Asus are the least ones with 3.5% whereas Oppo has 3.7%.

4.12 Os:-

- The bar chart shows the percentage of used devices out of all types of operating systems.

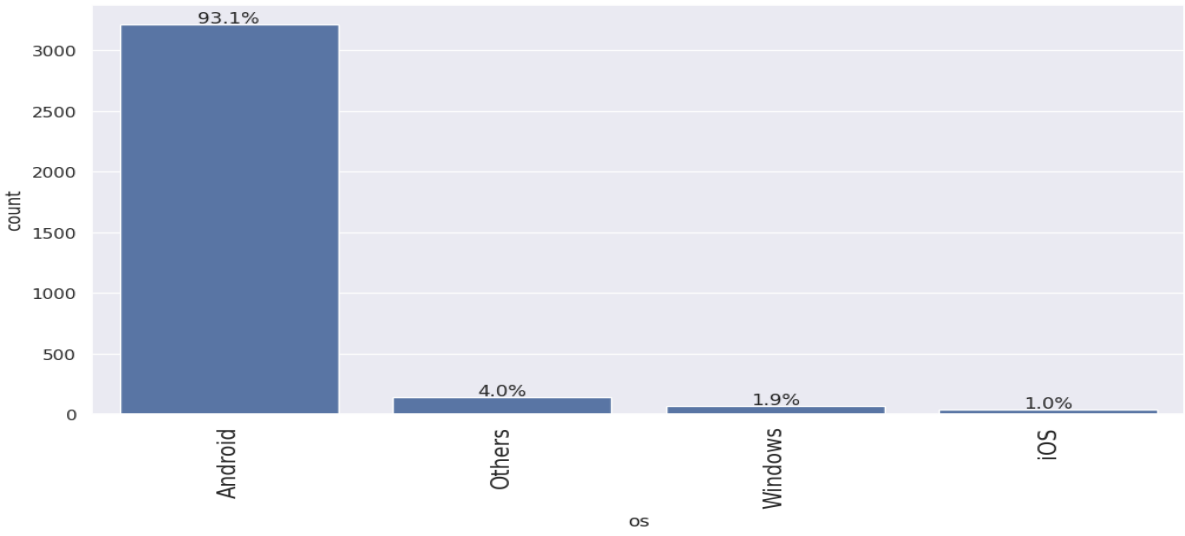

Figure 12 Os

- Overall 93.1 percent of the used devices market is dominated by Android devices.
- Windows Os is occupied by 1.9%, Ios is 1.0% & Others are 4.0%.

4.13 4g:-

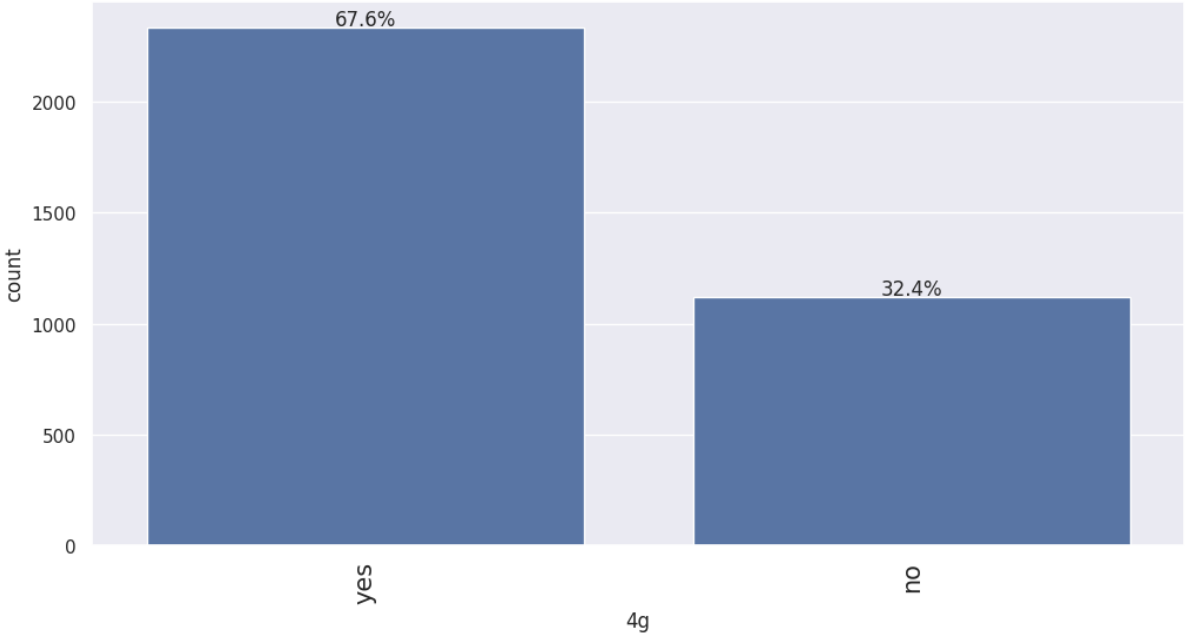

Figure 13 4G

- The bar chart shows whether the 4g is available or not in percentage.
- 4g network is available (i.e. yes) for 67.6% whereas 32.4% (i.e. no) is not available.

4.14 5g:-

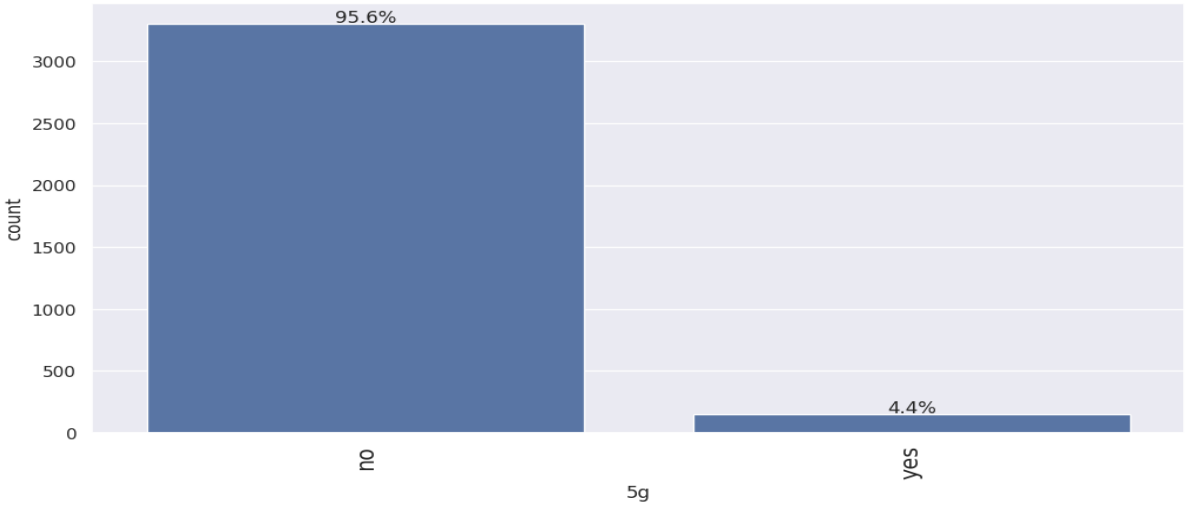

Figure 14 5G

- The bar chart shows in percent whether the 5g is available or not.
- 5g network is available (i.e. yes) for 95.6% whereas 4.4% (i.e. no) is not available.

4.15 release_year:- It means the Year when the device model was released

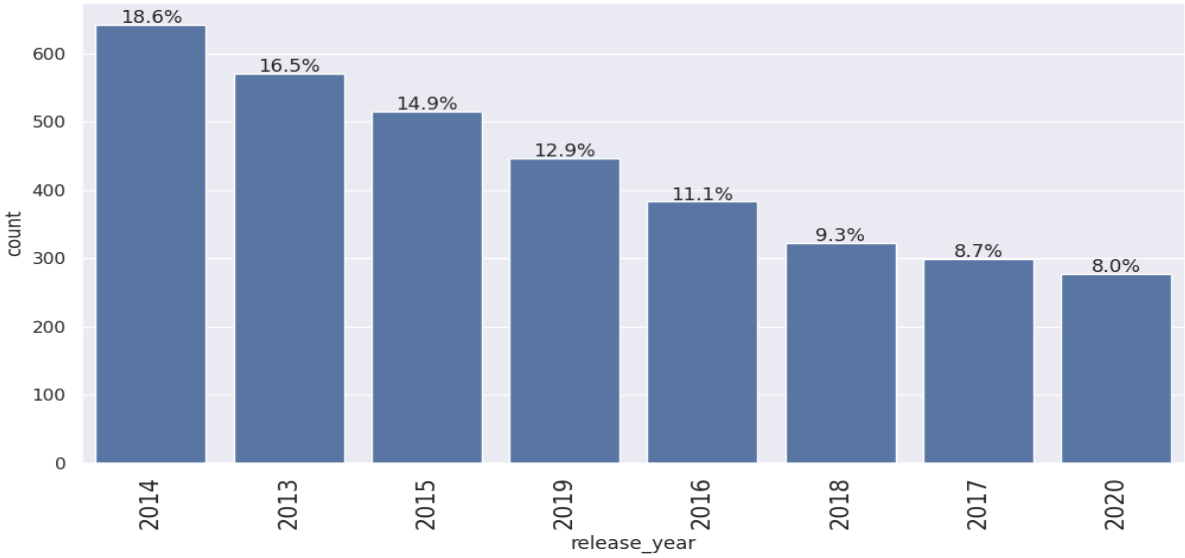

Figure 15 release_year

- In the year-2014 – 18.6 percent of the device models were released which is the highest rate.
- The percentage gradually decreased to 8.0% during 2020 which is the least portion.

5. Bivariate Analysis:-

5.1 Correlation Check:- we used a Heatmap to view the correlation between parameters.

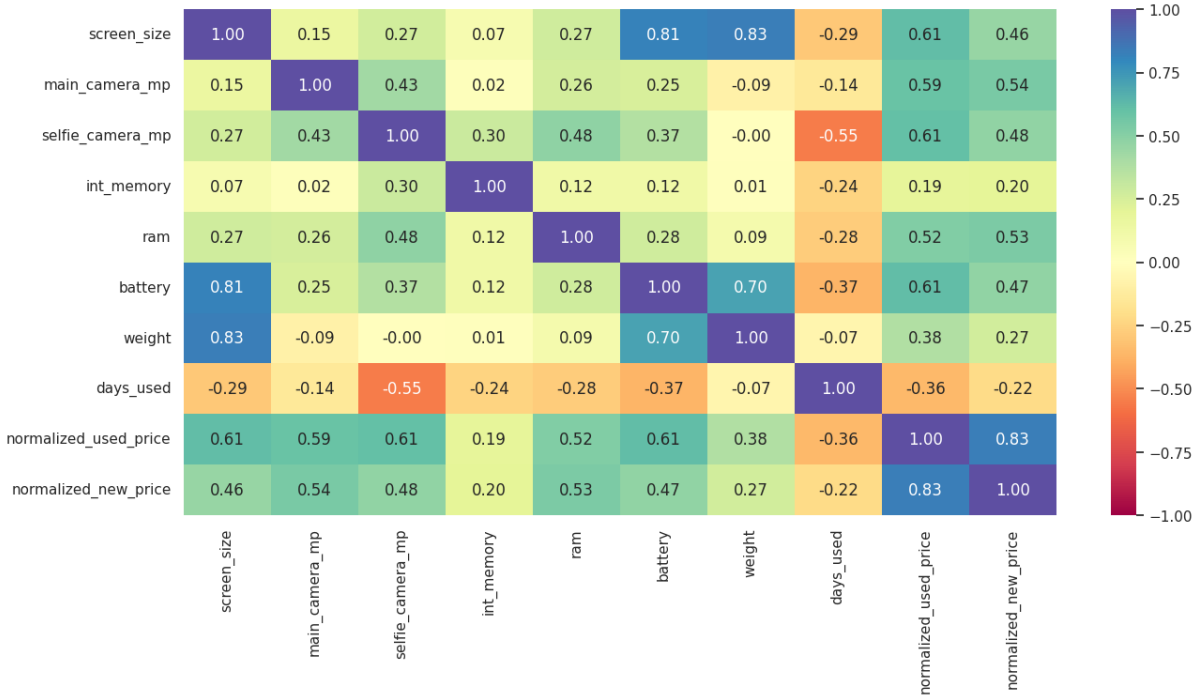

Figure 16 Correlation Check

- Noremalize_used_price, normalized_new_price, weight, and battery have a strong positive correlation.
- Selfie_camera_mp has a strong adverse correlation with days_used.
- Selfie_camera_mp does not correlate with weight.

5.2 The amount of RAM is important for the smooth functioning of a device. Let's see how the amount of RAM varies across brands:-

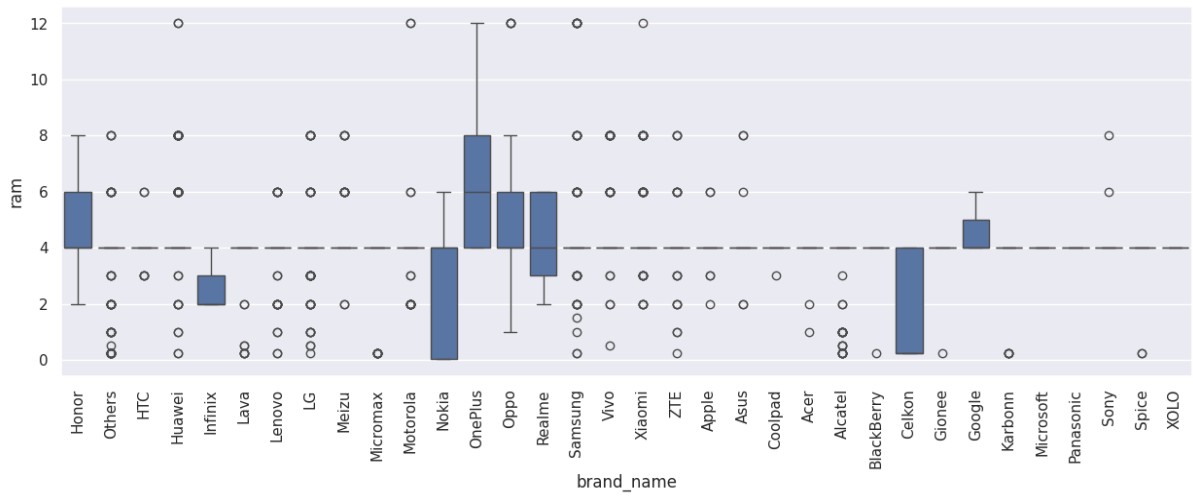

Figure 17 Amount of RAM varies across brands

- RAM under the brand name of OnePlus are rated the highest whereas those under the brand name of Nokia & Celkon are rated the least.

5.3 People who travel frequently require devices with large batteries to run through the day. But large battery often increases weight, making it feel uncomfortable in the hands. Let's create a new dataframe of only those devices which offer a large battery and analyze:-

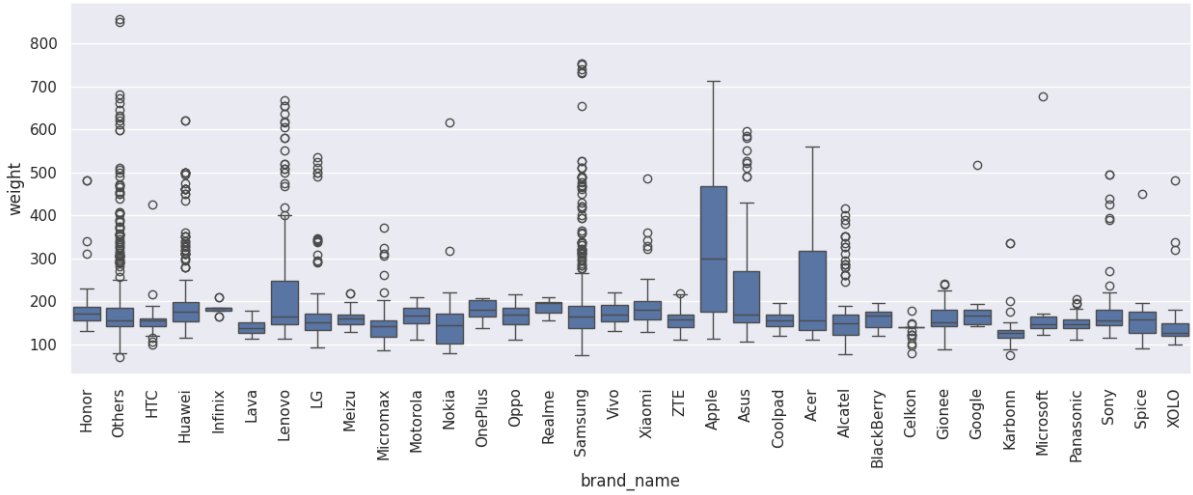

Figure 18 create a new dataframe of only those devices which offer a large battery and analyze

- The large batteries offering more than 4500 mAh are 341 devices across all the brand names.
- Frequent travelers often use Apple brand with large batteries (i.e. greater than 4500 mAh) so it offers the highest number of phones with large batteries. It has no outliers
- Karbonn brand is the last one to use with large batteries greater than 4500 mAh. It has 3 outliers.

5.4 People who buy phones and tablets primarily for entertainment purposes prefer a large screen as they offer a better viewing experience. Let's create a new dataframe of only those devices which are suitable for such people and analyze:-

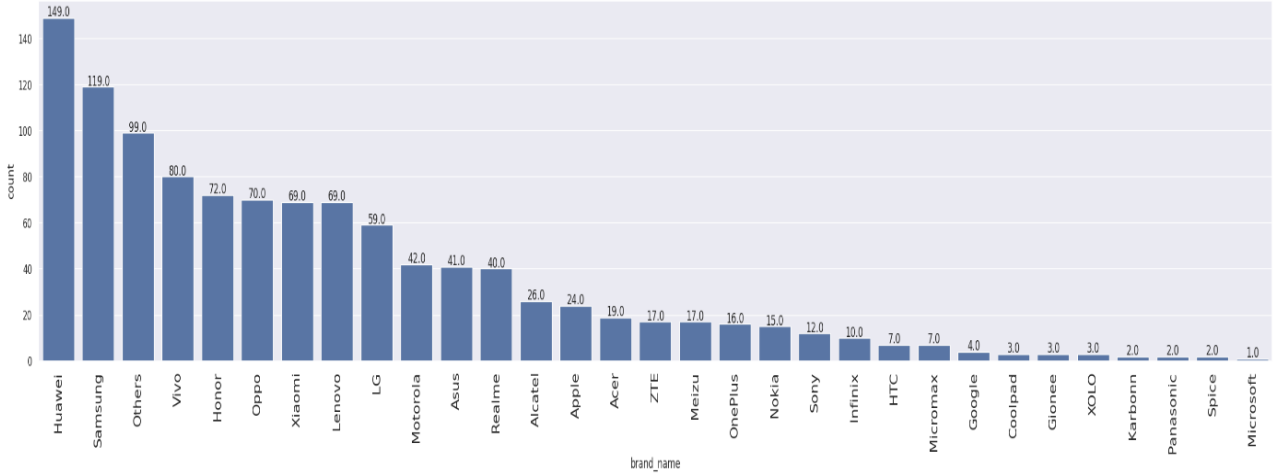

Figure 19 People who buy phones and tablets primarily for entertainment purposes prefer a large screen as they offer a better viewing experience. Let's create a new dataframe of only those devices which are suitable for such people and analyze

- The number of devices available with large screen sizes (i.e. screen sizes greater than 6 inches) is 1099 count out of all the brand name
- Huawei brand is offering the highest number of devices (149 phones/tablets) with large screens for entertainment purposes.
- The brand that offers the least count (1.0) of devices with large screens is Microsoft.

5.5 Everyone likes a good camera to capture their favorite moments with loved ones. Some customers specifically look for good front cameras to click cool selfies. Let's create a new dataframe of only those devices which are suitable for this customer segment and analyze:-

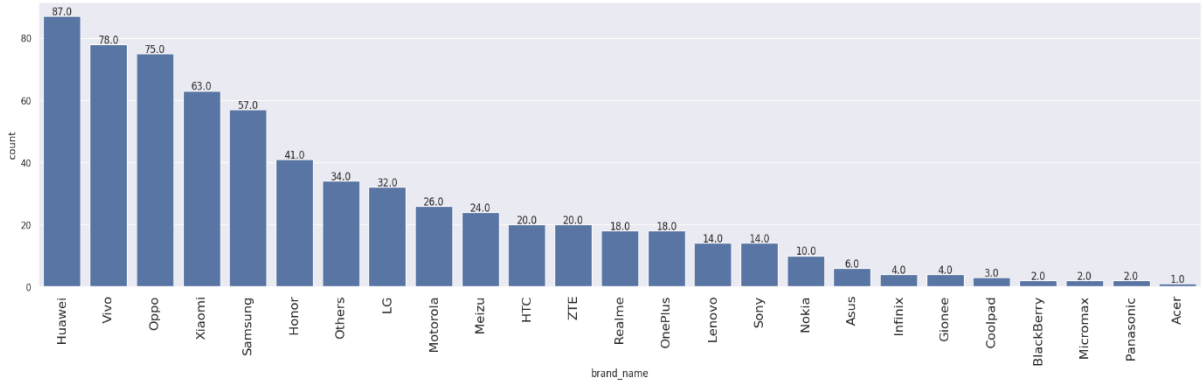

Figure 20 Everyone likes a good camera to capture their favorite moments with loved ones. Some customers specifically look for good front cameras to click cool selfies. Let's create a new dataframe of only those devices which are suitable for this customer segmen

- distribution of devices offering greater than 8MP selfie cameras across brands are 655 devices.
- 87 devices from Huawei brand are available with greater than 8MP selfie cameras which has been offered the highest count.
- 1.0 devices from Acer brand are available with greater than 8MP selfie cameras which has been offered the lower count.
- HTC & ZTE offers 20 devices and OnePlus & Realme with 18 devices in the middle.

5.6 Let's do a similar analysis for rear cameras.

Rear cameras generally have a better resolution than front cameras, so we set the threshold higher for them at 16MP.

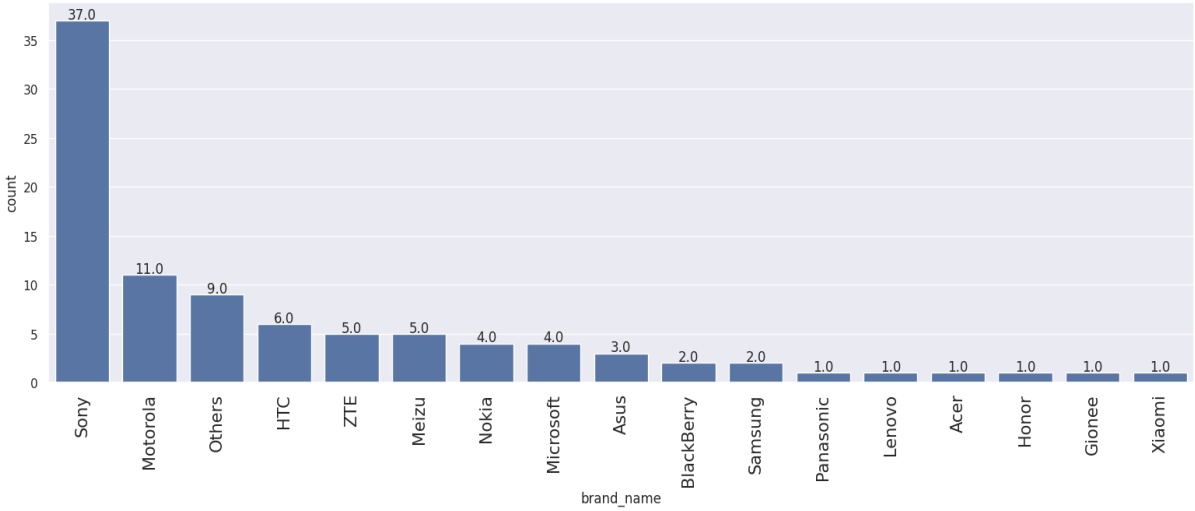

Figure 21 higher for them at 16MP

- distribution of devices offering greater than 16MP selfie cameras across brands are 94 devices.
- 37 devices from the Sony brand are available with greater than 16MP selfie cameras which has been offered the highest count.
- 1.0 devices from the Xiaomi brand are available with more than 16MP selfie cameras which has been offered the lowest number of devices.

5.7 Let's see how the price of used devices varies across the years.

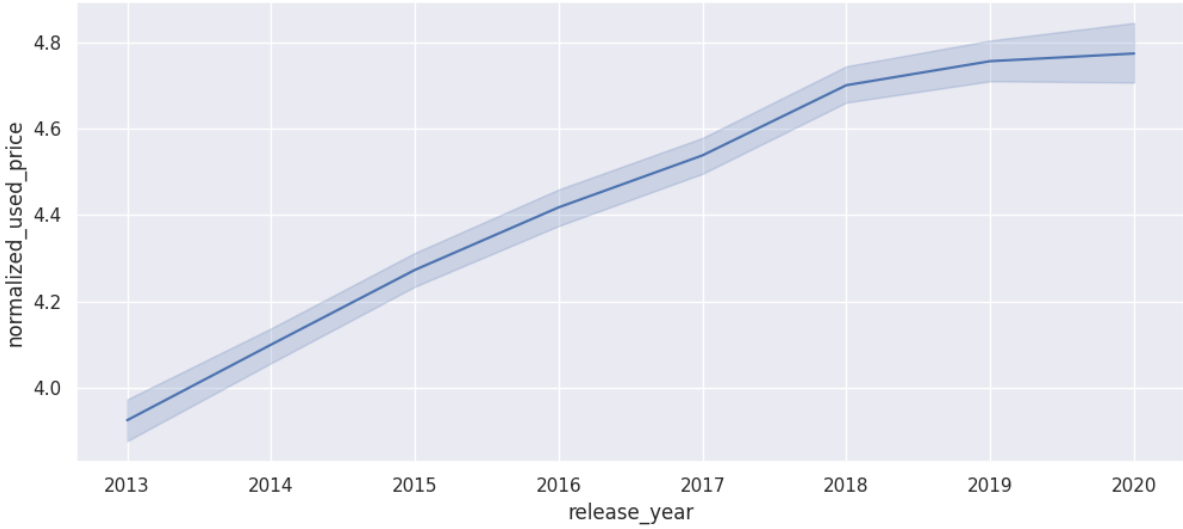

Figure 22 the price of used devices varies across the years

- The normalized used price increased gradually from the year 2013 to 2020.
- It is called positive correlation whereas the normalized used price increases along with release year.

5.8 Let's check how the prices vary for used phones and tablets offering 4G and 5G networks.

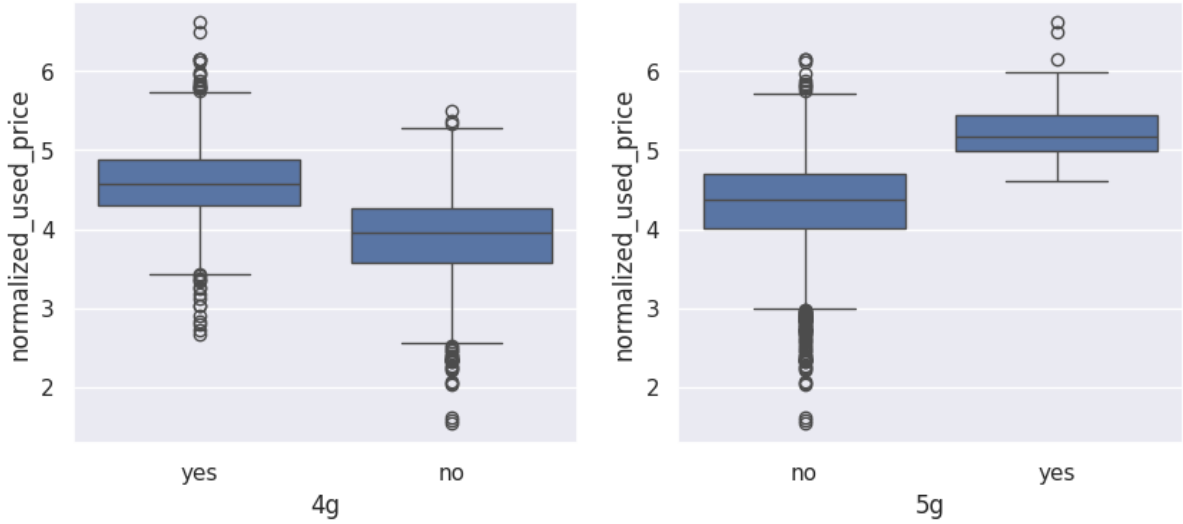

Figure 23 prices vary for used phones and tablets offering 4G and 5G networks

- The price of 5G networks (5.5) offered for used phones and tablets is higher than 4G networks (4.5).
- The price of no 5G networks (4.5) offered for used phones and tablets is higher than 4G networks (4).
- 5G has fewer Outliers in the case of “yes” and 4G has fewer outliers in the case of “no”.

6. Data Preprocessing:-

6.1 Missing Value Imputation

- We will impute the missing values in the data by the column medians grouped by release_year and brand_name.
- This table for missing values in all the columns.

	0
brand_name	0
os	0
screen_size	0
4g	0
5g	0
main_camera_mp	179
selfie_camera_mp	2
int_memory	4
ram	4
battery	6
weight	7
release_year	0
days_used	0
normalized_used_price	0
normalized_new_price	0

dtype: int64

Table 5 missing values in all the columns

We will impute the remaining missing values in the data by the column medians grouped by brand_name. Table for missing values after imputing the above columns.

	0
brand_name	0
os	0
screen_size	0
4g	0
5g	0
main_camera_mp	10
selfie_camera_mp	0
int_memory	0
ram	0
battery	0
weight	0
release_year	0
days_used	0
normalized_used_price	0
normalized_new_price	0

dtype: int64

Table 6 missing values after imputing brand name

We will fill the remaining missing values in the main_camera_mp column by the column median.

The table for missing values after imputing the above columns.

	0
brand_name	0
os	0
screen_size	0
4g	0
5g	0
main_camera_mp	0
selfie_camera_mp	0
int_memory	0
ram	0
battery	0
weight	0
release_year	0
days_used	0
normalized_used_price	0
normalized_new_price	0

dtype: int64

Table 7 missing values after imputing main camera

6.2 Feature Engineering:-

- Let's create a new column years_since_release from the release_year column.
- We will consider the year of data collection, 2021, as the baseline.
- We will drop the release_year column and this table for a statistical summary.

	years_since_release
count	3454.000000
mean	5.034742
std	2.298455
min	1.000000
25%	3.000000
50%	5.500000
75%	7.000000
max	8.000000

dtype: float64

Table 8 drop the release_year column and this table for a statistical summary

6.3 Outlier Check:- Let's check for outliers in the data using boxplot.

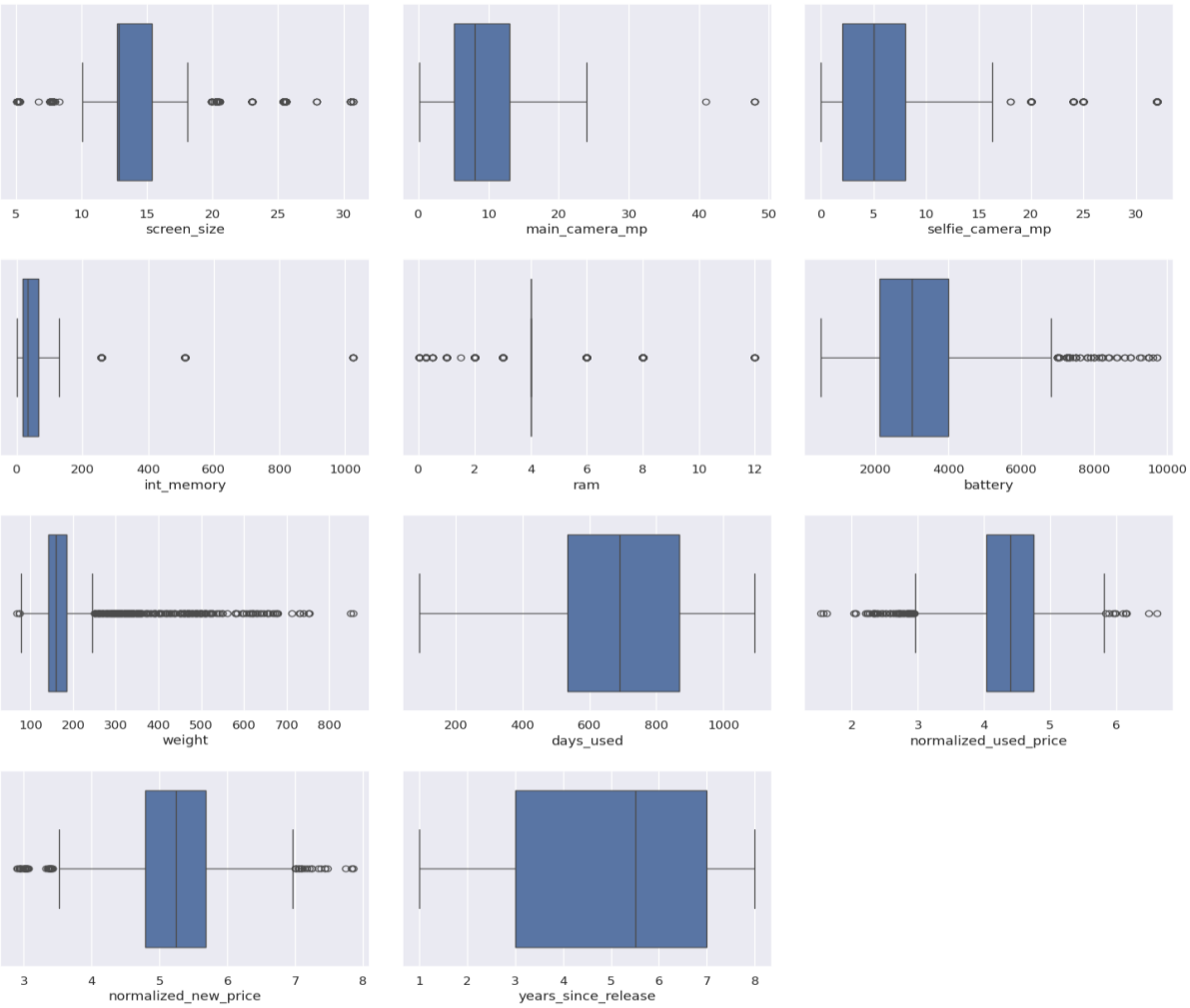

Figure 24 Outlier Check

- Main camera mp, selfie camera mp, int memory, ram, battery, weight, normalized used price, have outliers on the right-hand side so it has more distance between Q3 Quartile & Maximum.
- Days used & years since release have no outliers so it gives genuine results.

6.4 Data Preparation for Modelling:-

We want to predict the normalized price of used devices.

Before we proceed to build a model, we'll have to encode categorical features.

We'll split the data into train and test to be able to evaluate the model that we build on the train data.

We will build a Linear Regression model using the train data and then check its performance

The below table defines the dependent and independent variables.

```
  brand_name       os  screen_size   4g   5g  main_camera_mp  \
0      Honor  Android        14.50  yes   no            13.0
1      Honor  Android        17.30  yes  yes            13.0
2      Honor  Android        16.69  yes  yes            13.0
3      Honor  Android        25.50  yes  yes            13.0
4      Honor  Android        15.32  yes   no            13.0

   selfie_camera_mp  int_memory  ram  battery  weight  days_used  \
0               5.0        64.0  3.0   3020.0   146.0        127
1              16.0       128.0  8.0   4300.0   213.0        325
2               8.0       128.0  8.0   4200.0   213.0        162
3               8.0        64.0  6.0   7250.0   480.0        345
4               8.0        64.0  3.0   5000.0   185.0        293

   normalized_new_price  years_since_release
0              4.715100                    1
1              5.519018                    1
2              5.884631                    1
3              5.630961                    1
4              4.947837                    1

0    4.307572
1    5.162097
2    5.111084
3    5.135387
4    4.389995
Name: normalized_used_price, dtype: float64
```

Table 9 Data Preparation for Modelling

creating dummy variables: - This is to create dummies for independent features.

The Number of rows is 5 and the number of columns is 49.

	const	screen_size	main_camera_mp	selfie_camera_mp	int_memory	ram	battery	weight	days_used	normalized_new_price	...	brand_name_Spice	brand_name_Vivo	brand_name_XOLO	brand_name_Xiaomi	brand_name_ZTE	os_Others	os_Windows	os_iOS	4g_yes	5g_yes
0	1.0	14.50	13.0	5.0	64.0	3.0	3020.0	146.0	127	4.715100	...	False	False	False	False	False	False	False	False	True	False
1	1.0	17.30	13.0	16.0	128.0	8.0	4300.0	213.0	325	5.519018	...	False	False	False	False	False	False	False	False	True	True
2	1.0	16.69	13.0	8.0	128.0	8.0	4200.0	213.0	162	5.884631	...	False	False	False	False	False	False	False	False	True	True
3	1.0	25.50	13.0	8.0	64.0	6.0	7250.0	480.0	345	5.630961	...	False	False	False	False	False	False	False	False	True	True
4	1.0	15.32	13.0	8.0	64.0	3.0	5000.0	185.0	293	4.947837	...	False	False	False	False	False	False	False	False	True	False

5 rows × 49 columns

Table 10 Number of rows is 5 and the number of columns is 49

Splitting the data in a 70:30 ratio for train-to-test data: -

The number of rows in train data = 2417 as seen in 70% ratio

Number of rows in test data = 1037 as seen 30% ratio

7. Model Building- Linear Regression

```
                            OLS Regression Results
==============================================================================
Dep. Variable:     normalized_used_price   R-squared:                       0.845
Model:                               OLS   Adj. R-squared:                  0.842
Method:                    Least Squares   F-statistic:                     268.7
Date:                   Mon, 02 Sep 2024   Prob (F-statistic):               0.00
Time:                           14:21:43   Log-Likelihood:                 123.85
No. Observations:                   2417   AIC:                            -149.7
Df Residuals:                       2368   BIC:                             134.0
Df Model:                             48
Covariance Type:                nonrobust
==========================================================================================
                             coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------------------
const                      1.3156      0.071     18.454      0.000       1.176       1.455
screen_size                0.0244      0.003      7.163      0.000       0.018       0.031
main_camera_mp             0.0208      0.002     13.848      0.000       0.018       0.024
selfie_camera_mp           0.0135      0.001     11.997      0.000       0.011       0.016
int_memory                 0.0001   6.97e-05      1.651      0.099   -2.16e-05       0.000
ram                        0.0230      0.005      4.451      0.000       0.013       0.033
battery                -1.689e-05   7.27e-06     -2.321      0.020   -3.12e-05   -2.62e-06
weight                     0.0010      0.000      7.480      0.000       0.001       0.001
days_used               4.216e-05   3.09e-05      1.366      0.172   -1.84e-05       0.000
normalized_new_price       0.4311      0.012     35.147      0.000       0.407       0.455
years_since_release       -0.0237      0.005     -5.193      0.000      -0.033      -0.015
brand_name_Alcatel         0.0154      0.048      0.323      0.747      -0.078       0.109
brand_name_Apple          -0.0038      0.147     -0.026      0.980      -0.292       0.285
brand_name_Asus            0.0151      0.048      0.314      0.753      -0.079       0.109
brand_name_BlackBerry     -0.0300      0.070     -0.427      0.669      -0.168       0.108
brand_name_Celkon         -0.0468      0.066     -0.707      0.480      -0.177       0.083
brand_name_Coolpad         0.0209      0.073      0.287      0.774      -0.122       0.164
brand_name_Gionee          0.0448      0.058      0.775      0.438      -0.068       0.158
brand_name_Google         -0.0326      0.085     -0.385      0.700      -0.199       0.133
brand_name_HTC            -0.0130      0.048     -0.270      0.787      -0.108       0.081
brand_name_Honor           0.0317      0.049      0.644      0.520      -0.065       0.128
brand_name_Huawei         -0.0020      0.044     -0.046      0.964      -0.089       0.085
brand_name_Infinix         0.1633      0.093      1.752      0.080      -0.019       0.346
brand_name_Karbonn         0.0943      0.067      1.405      0.160      -0.037       0.226
brand_name_LG             -0.0132      0.045     -0.291      0.771      -0.102       0.076
brand_name_Lava            0.0332      0.062      0.533      0.594      -0.089       0.155
brand_name_Lenovo          0.0454      0.045      1.004      0.316      -0.043       0.134
brand_name_Meizu          -0.0129      0.056     -0.230      0.818      -0.123       0.097
brand_name_Micromax       -0.0337      0.048     -0.704      0.481      -0.128       0.060
brand_name_Microsoft       0.0952      0.088      1.078      0.281      -0.078       0.268
brand_name_Motorola       -0.0112      0.050     -0.226      0.821      -0.109       0.086
brand_name_Nokia           0.0719      0.052      1.387      0.166      -0.030       0.174
brand_name_OnePlus         0.0709      0.077      0.916      0.360      -0.081       0.223
brand_name_Oppo            0.0124      0.048      0.261      0.794      -0.081       0.106
brand_name_Others         -0.0080      0.042     -0.190      0.849      -0.091       0.075
brand_name_Panasonic       0.0563      0.056      1.008      0.314      -0.053       0.166
brand_name_Realme          0.0319      0.062      0.518      0.605      -0.089       0.153
brand_name_Samsung        -0.0313      0.043     -0.725      0.469      -0.116       0.053
brand_name_Sony           -0.0616      0.050     -1.220      0.223      -0.161       0.037
brand_name_Spice          -0.0147      0.063     -0.233      0.816      -0.139       0.109
brand_name_Vivo           -0.0154      0.048     -0.318      0.750      -0.110       0.080
brand_name_XOLO            0.0152      0.055      0.277      0.782      -0.092       0.123
brand_name_Xiaomi          0.0869      0.048      1.806      0.071      -0.007       0.181
brand_name_ZTE            -0.0057      0.047     -0.121      0.904      -0.099       0.087
os_Others                 -0.0510      0.033     -1.555      0.120      -0.115       0.013
os_Windows                -0.0207      0.045     -0.459      0.646      -0.109       0.068
os_iOS                    -0.0663      0.146     -0.453      0.651      -0.354       0.221
4g_yes                     0.0528      0.016      3.326      0.001       0.022       0.084
5g_yes                    -0.0714      0.031     -2.268      0.023      -0.133      -0.010
==============================================================================
Omnibus:                      223.612   Durbin-Watson:                   1.910
Prob(Omnibus):                  0.000   Jarque-Bera (JB):              422.275
Skew:                          -0.620   Prob(JB):                     2.01e-92
Kurtosis:                       4.630   Cond. No.                     1.78e+05
==============================================================================
```

Table 11 Model Building - Linear Regression

Notes: -

- Standard Errors assume that the covariance matrix of the errors is correctly specified.
- The condition number is large, 1.78e+05. This might indicate that there are strong multicollinearity or other numerical problems.

7.1 Model Performance Check:-

Let's check the performance of the model using different metrics.

- We will be using metric functions defined in sklearn for RMSE, MAE, and
- We will define a function to calculate MAPE and adjusted
- We will create a function which will print out all the above metrics in one go.

Training Performance is 70% data

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	0.229884	0.180326	0.844886	0.841675	4.326841

Table 12 Training Performance is 70% data

The regression model has a strong performance with:

- RMSE & MAE are Low average errors.
- R-squared and Adjusted R-squared are a good fit for our data.
- MAPE indicates a 4.3% of predictive accuracy.

Test Performance as 30% data

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	0.238358	0.184749	0.842479	0.834659	4.501651

Table 13 Test Performance as 30% data

- The values suggest our regression model is a relatively good fit for our data as it has high R-squared and Adj.R-squared values.
- RMSE & MAE are relatively low which means predictions are fairly close to the actual values.
- MAPE indicates a 4.5% average absolute percentage error which can be acceptable.

8. Checking Linear Regression Assumptions

We will be checking the following Linear Regression assumptions:

- No Multicollinearity
- Linearity of variables
- Independence of error terms
- Normality of error terms
- No Heteroscedasticity

8.1 TEST FOR MULTICOLLINEARITY:-

We will test for multicollinearity using VIF.

General Rule of thumb:

- If VIF is 1 then there is no correlation between the predictor and the remaining predictor variables.
- If VIF exceeds 5 or is close to exceeding 5, we say there is moderate multicollinearity.
- If VIF is 10 or exceeding 10, it shows signs of high multicollinearity.

Let's define a function to check VIF & check VIF on train data.

Table 14 defines a function to check VIF

Sl.No.	Feature	VIF
0	const	227.744081
1	screen_size	7.67729
2	main_camera_mp	2.285051
3	selfie_camera_mp	2.812473
4	int_memory	1.364152
5	ram	2.282352
6	battery	4.08178
7	weight	6.396749
8	days_used	2.660269
9	normalized_new_price	3.11943
10	years_since_release	4.899007
11	brand_name_Alcatel	3.405693
12	brand_name_Apple	13.057668
13	brand_name_Asus	3.332038
14	brand_name_BlackBerry	1.632378
15	brand_name_Celkon	1.774721
16	brand_name_Coolpad	1.468006
17	brand_name_Gionee	1.951272
18	brand_name_Google	1.321778
19	brand_name_HTC	3.410361
20	brand_name_Honor	3.340687
21	brand_name_Huawei	5.983852
22	brand_name_Infinix	1.283955
23	brand_name_Karbonn	1.573702
24	brand_name_LG	4.849832
25	brand_name_Lava	1.71136
26	brand_name_Lenovo	4.558941
27	brand_name_Meizu	2.179607
28	brand_name_Micromax	3.363521
29	brand_name_Microsoft	1.869751
30	brand_name_Motorola	3.274558
31	brand_name_Nokia	3.479849
32	brand_name_OnePlus	1.437034
33	brand_name_Oppo	3.971194
34	brand_name_Others	9.711034
35	brand_name_Panasonic	2.105703
36	brand_name_Realme	1.946812
37	brand_name_Samsung	7.539866
38	brand_name_Sony	2.943161
39	brand_name_Spice	1.688863
40	brand_name_Vivo	3.651437
41	brand_name_XOLO	2.13807
42	brand_name_Xiaomi	3.719689
43	brand_name_ZTE	3.797581

44	os_Others	1.859863
45	os_Windows	1.596034
46	os_iOS	11.784684
47	4g_yes	2.467681
48	5g_yes	1.8139

8.2 Removing Multicollinearity (if needed)

To remove multicollinearity

1. Drop every column one by one that has a VIF score greater than 5.
2. Look at the adjusted R-squared and RMSE of all these models.
3. Drop the variable that makes the least change in the adjusted R-squared.
4. Check the VIF scores again.
5. Continue till you get all VIF scores under 5.

Let's define a function that will help us do this.

Specify the columns with high VIF: -

	col	Adj. R-squared after_dropping col	RMSE after dropping col
0	brand_name_Apple	0.841809	0.232201
1	os_iOS	0.841795	0.232211

VIF scores after dropping os_iOS

Sl.No.	feature	VIF
0	const	226.533454
1	screen_size	7.608413
2	main_camera_mp	2.282357
3	selfie_camera_mp	2.802123
4	int_memory	1.364151
5	ram	2.266773
6	battery	4.077873
7	weight	6.378393
8	days_used	2.659991
9	normalized_new_price	3.119423
10	years_since_release	4.897569
11	brand_name_Alcatel	3.40567
12	brand_name_Apple	2.114776
13	brand_name_Asus	3.332036
14	brand_name_BlackBerry	1.629452
15	brand_name_Celkon	1.774714
16	brand_name_Coolpad	1.467927
17	brand_name_Gionee	1.9511
18	brand_name_Google	1.321735
19	brand_name_HTC	3.40984

20	brand_name_Honor	3.340579
21	brand_name_Huawei	5.983729
22	brand_name_Infinix	1.283954
23	brand_name_Karbonn	1.573287
24	brand_name_LG	4.849466
25	brand_name_Lava	1.710982
26	brand_name_Lenovo	4.558917
27	brand_name_Meizu	2.179601
28	brand_name_Micromax	3.363328
29	brand_name_Microsoft	1.869578
30	brand_name_Motorola	3.274541
31	brand_name_Nokia	3.476718
32	brand_name_OnePlus	1.436973
33	brand_name_Oppo	3.970991
34	brand_name_Others	9.710594
35	brand_name_Panasonic	2.105578
36	brand_name_Realme	1.946739
37	brand_name_Samsung	7.539698
38	brand_name_Sony	2.943107
39	brand_name_Spice	1.688086
40	brand_name_Vivo	3.65141
41	brand_name_XOLO	2.137798
42	brand_name_Xiaomi	3.719646
43	brand_name_ZTE	3.797186
44	os_Others	1.746926
45	os_Windows	1.59437
46	4g_yes	2.466649
47	5g_yes	1.808206

Table 15 VIF scores after dropping os_iOS

8.3 Dropping high p-value variables (if needed)

- We will drop the predictor variables having a p-value greater than 0.05 as they do not significantly impact the target variable.
- But sometimes p-values change after dropping a variable. So, we'll not drop all variables at once.
- Instead, we will do the following:
    - Build a model, check the p-values of the variables, and drop the column with the highest p-value.
    - Create a new model without the dropped feature, check the p-values of the variables, and drop the column with the highest p-value.
    - Repeat the above two steps till there are no columns with p-value > 0.05.

The above process can also be done manually by picking one variable at a time that has a high p-value, dropping it, and building a model again. But that might be a little tedious and using a loop will be more efficient.

Selected_features (Cols)

['const', 'screen_size', 'main_camera_mp', 'selfie_camera_mp', 'ram', 'battery', 'weight', 'normalized_new_price', 'years_since_release', 'brand_name_Lenovo', 'brand_name_Nokia', 'brand_name_Xiaomi', 'os_Others', '4g_yes']

```
                            OLS Regression Results
==============================================================================
Dep. Variable:     normalized_used_price   R-squared:                       0.842
Model:                               OLS   Adj. R-squared:                  0.842
Method:                    Least Squares   F-statistic:                     988.1
Date:                   Mon, 02 Sep 2024   Prob (F-statistic):               0.00
Time:                           16:12:30   Log-Likelihood:                 104.71
No. Observations:                   2417   AIC:                            -181.4
Df Residuals:                       2403   BIC:                            -100.4
Df Model:                             13
Covariance Type:               nonrobust
==========================================================================================
                           coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------------------
const                    1.3777      0.051     26.879      0.000       1.277       1.478
screen_size              0.0256      0.003      7.764      0.000       0.019       0.032
main_camera_mp           0.0212      0.001     15.313      0.000       0.018       0.024
selfie_camera_mp         0.0140      0.001     13.203      0.000       0.012       0.016
ram                      0.0175      0.004      3.950      0.000       0.009       0.026
battery              -1.507e-05    7.1e-06     -2.122      0.034    -2.9e-05   -1.14e-06
weight                   0.0009      0.000      7.177      0.000       0.001       0.001
normalized_new_price     0.4222      0.011     39.125      0.000       0.401       0.443
years_since_release     -0.0199      0.004     -5.516      0.000      -0.027      -0.013
brand_name_Lenovo        0.0492      0.021      2.288      0.022       0.007       0.091
brand_name_Nokia         0.0675      0.031      2.203      0.028       0.007       0.128
brand_name_Xiaomi        0.0893      0.026      3.498      0.000       0.039       0.139
os_Others               -0.0704      0.030     -2.356      0.019      -0.129      -0.012
4g_yes                   0.0499      0.015      3.357      0.001       0.021       0.079
==============================================================================
Omnibus:                      232.847   Durbin-Watson:                   1.913
Prob(Omnibus):                  0.000   Jarque-Bera (JB):              458.097
Skew:                          -0.628   Prob(JB):                    3.35e-100
Kurtosis:                       4.724   Cond. No.                     3.85e+04
==============================================================================
```

Table 16 OLS Regression results

Notes: -

- Standard Errors assume that the covariance matrix of the errors is correctly specified.
- The condition number is large, 3.85e+04. This might indicate that there are strong multicollinearity or other numerical problems.

Checking model performance on train set as 70% data

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	0.231712	0.181763	0.842409	0.841491	4.359962

Table 17 model performance on train set as 70% data

checking model performance on test set (seen 30% data)

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	0.236807	0.18348	0.844523	0.842393	4.467055

Table 18 model performance on test set (seen 30% data)

Now we'll check the rest of the assumptions on olsmod2: -

2. Linearity of variables
3. Independence of error terms
4. Normality of error terms
5. No Heteroscedasticity

8.4 TEST FOR LINEARITY AND INDEPENDENCE

- We will test for linearity and independence by making a plot of fitted values vs residuals and checking for patterns.
- If there is no pattern, then we say the model is linear and residuals are independent.
- Otherwise, the model is showing signs of non-linearity and residuals are not independent.

	Actual Values	Fitted Values	Residuals
3026	4.087488	3.855243	0.232245
1525	4.448399	4.628813	-0.180413
1128	4.315353	4.280778	0.034574
3003	4.282068	4.206270	0.075798
2907	4.456438	4.470315	-0.013877

Table 19 fitted values vs residuals

let's plot the fitted values vs residuals

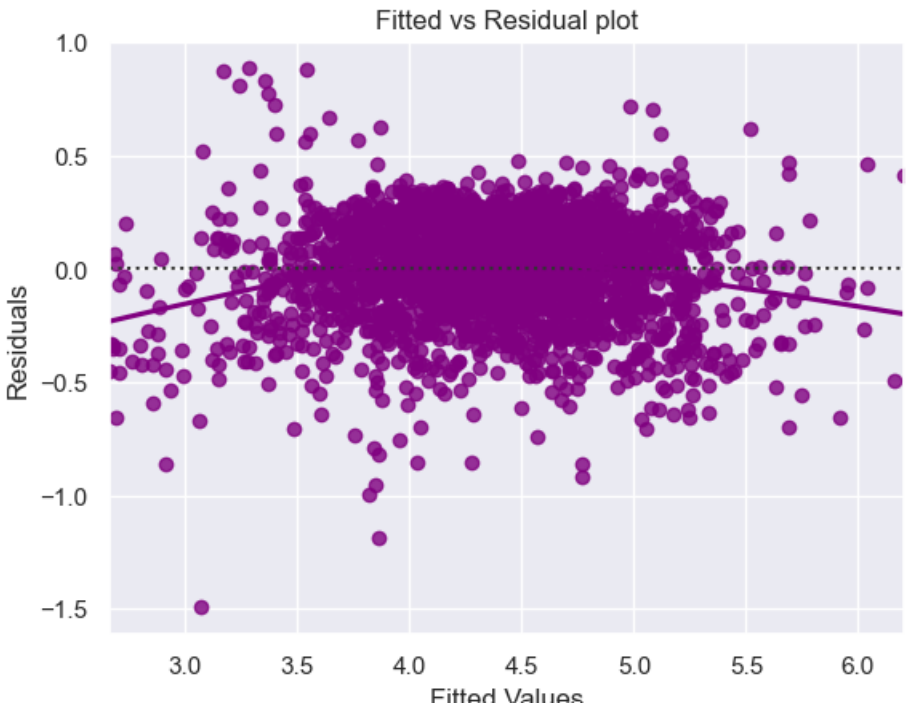

Figure 25 plot the fitted values vs residuals

8.5 TEST FOR NORMALITY

- We will test for normality by checking the distribution of residuals, by checking the Q-Q plot of residuals, and by using the Shapiro-Wilk test.
- If the residuals follow a normal distribution, they will make a straight-line plot, otherwise not.
- If the p-value of the Shapiro-Wilk test is greater than 0.05, we can say the residuals are normally distributed.

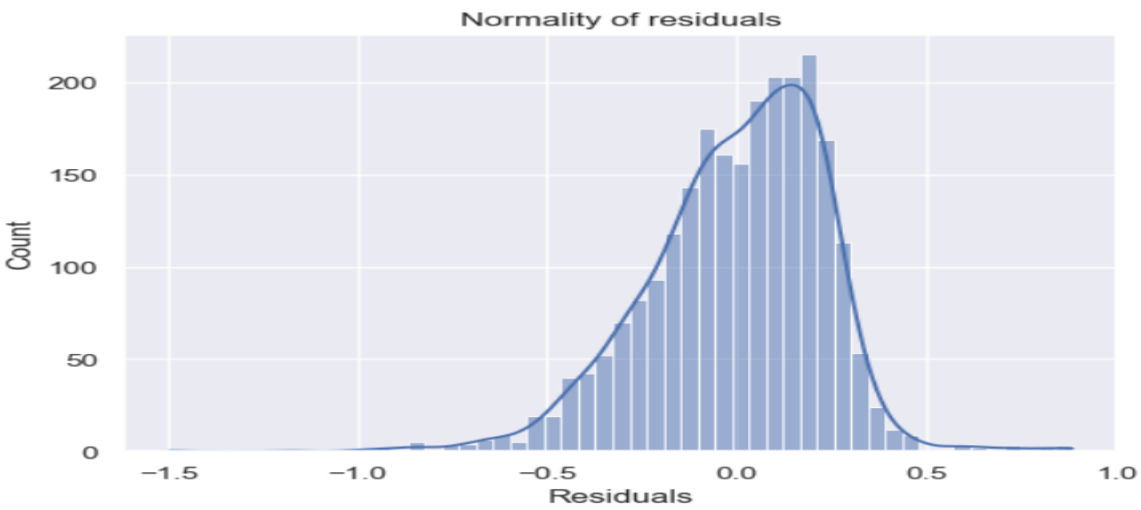

Figure 26 normality by checking the distribution of residuals

Check the Q-Q plot

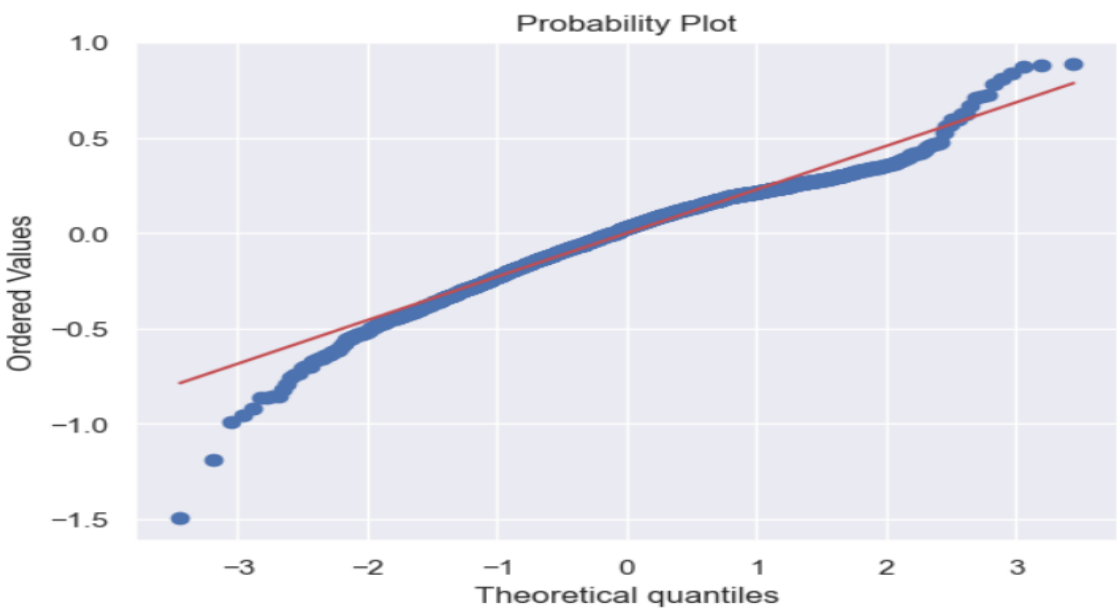

Figure 27 checking the Q-Q plot of residuals

ShapiroResult(statistic=0.9690964221954346, pvalue=2.131064442768085e-22)

Note: -

- This is normally distributed residual p-value of the Shapiro-Wilk test is greater than 0.05.

8.6 TEST FOR HOMOSCEDASTICITY

- We will test for homoscedasticity by using the Goldfeldquandt test.
- If we get a p-value greater than 0.05, we can say that the residuals are homoscedastic. Otherwise, they are heteroscedastic.

[('F statistic', 1.0438035947010254), ('p-value', 0.22944475832467995)]

Note: -

- The residuals are homoscedastic as the p-value is greater than 0.05.

9. Final Model Summary

Complete the code to fit the final model

```
                            OLS Regression Results
==============================================================================
Dep. Variable:     normalized_used_price   R-squared:                       0.842
Model:                               OLS   Adj. R-squared:                  0.842
Method:                    Least Squares   F-statistic:                     988.1
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const                    1.3777      0.051     26.879      0.000       1.277       1.478
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main_camera_mp           0.0212      0.001     15.313      0.000       0.018       0.024
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weight                   0.0009      0.000      7.177      0.000       0.001       0.001
normalized_new_price     0.4222      0.011     39.125      0.000       0.401       0.443
years_since_release     -0.0199      0.004     -5.516      0.000      -0.027      -0.013
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brand_name_Nokia         0.0675      0.031      2.203      0.028       0.007       0.128
brand_name_Xiaomi        0.0893      0.026      3.498      0.000       0.039       0.139
os_Others               -0.0704      0.030     -2.356      0.019      -0.129      -0.012
4g_yes                   0.0499      0.015      3.357      0.001       0.021       0.079
==============================================================================
Omnibus:                      232.847   Durbin-Watson:                   1.913
Prob(Omnibus):                  0.000   Jarque-Bera (JB):              458.097
Skew:                          -0.628   Prob(JB):                    3.35e-100
Kurtosis:                       4.724   Cond. No.                     3.85e+04
==============================================================================
```

Table 20 Final Model Summary

Notes: -

- Standard Errors assume that the covariance matrix of the errors is correctly specified.
- The condition number is large, 3.85e+04. This might indicate that there are strong multicollinearity or other numerical problems.

Checking model performance on train set (seen 70% data)

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	0.231712	0.181763	0.842409	0.841491	4.359962

Table 21 model performance on train set (seen 70% data)

Checking model performance on test set (seen 30% data)

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	0.236807	0.18348	0.844523	0.842393	4.467055

Table 22 model performance on test set (seen 30% data)

The small differences between training and test performance metrics demonstrate that the model is a good fit to perform consistently across datasets, making it a desirable outcome.

10. Actionable Insights and Recommendations:-

10.1 Actionable Insights:-

Overall, the below findings are identified and either directly/indirectly impacting impact the Foodhub restaurant business. So given advices as well.

- 7 devices are missed to show weights under the weight column.
- Only 5G is available in a few devices wherever it is possible to add 4G as well.
- Normalized used price and new price have Outliers which has more distance between Q3 & Maximum
- Most of the cases outliers are more than the maximum like screen size, main camera mp, selfie camera mp etc.
- Most of the devices are inbuilt with Android which may leads to monopoly in the market.
- Device models released were decreased from 2014 to 2020 which indicates reducing trend.
- Linear Regression assumptions contains higher amount os VIF.

10.2 Recommendations:-

All these recommendations will increase the foodhub restaurant business growth on day-to-day operations.

- It needs to add some weights for those 7 devices.
- Include both 4G & 5G services wherever only one of them is available.
- Reduce outliers to increase the effectiveness of the price.
- Reduce those outliers which are greater than maximum like screen size, main camera mp, selfie camera mp etc. so that will reduce the skewness and increase to distribute the values normally.
- Minimize the Monopoly power by offering the other Os as well.
- Better to update the models with new features to increase release in the device models.

- Control the high value of VIF to make the linear regression model as good fit.